

Fourteenth Section. Groups of Linear Ternary Substitutions.

§ 115. A ternary linear substitution group of order 168.

The question of the totality of all possible finite groups of linear substitutions in several dimensions, in particular in three and four dimensions, was treated by C. Jordan, who, in the same way as for binary substitutions, set up a finite number of types of such groups.¹ We cannot pursue these general investigations here; we content ourselves with considering one example of a ternary group more fully.

In the section on congruence groups we became acquainted with a series of simple groups. The first, of order 60, proved to be isomorphic with the icosahedral group. We now consider the next of these simple groups, of order 168, denoted there by L_7 , and try to construct a ternary linear substitution group isomorphic with it. The theorem of § 72 leads to this construction once we find four elements $\tau, \chi, \theta, \omega$ whose products satisfy the relations stated there.

We first take

$$\tau = \begin{pmatrix} \varepsilon & 0 & 0 \\ 0 & \varepsilon^2 & 0 \\ 0 & 0 & \varepsilon^4 \end{pmatrix}, \quad \chi = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \chi^3 = 1, \quad (1)$$

where ε is an imaginary seventh root of unity. The determinant of τ is one, since

$$\varepsilon \varepsilon^2 \varepsilon^4 = 1. \quad (2)$$

The two substitutions τ and χ already generate a group of order 21. The form of χ is that of a cyclic permutation of the three variables.

To determine the further substitution ω , we take it in the circulant form

$$\omega = \begin{pmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{pmatrix}. \quad (3)$$

The condition that ω be of second order and have determinant one gives the relations

$$\alpha^2 + 2\beta\gamma = 1, \quad \beta^2 + 2\gamma\alpha = 0, \quad \gamma^2 + 2\alpha\beta = 0, \quad (4)$$

$$\alpha + \beta + \gamma = -1. \quad (5)$$

From these follow

$$\alpha = \beta\gamma - \alpha^2, \quad \beta = \gamma\alpha - \beta^2, \quad \gamma = \alpha\beta - \gamma^2, \quad (6)$$

and

$$\alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma = -1. \quad (7)$$

Comparison of the remaining required relations yields

$$\alpha = h(\varepsilon^4 - \varepsilon^{-4}), \quad \beta = h(\varepsilon^2 - \varepsilon^{-2}), \quad \gamma = h(\varepsilon - \varepsilon^{-1}), \quad (8)$$

where

$$-1 = h(\varepsilon + \varepsilon^2 + \varepsilon^4 - \varepsilon^{-1} - \varepsilon^{-2} - \varepsilon^{-4}), \quad h = \frac{1}{\sqrt{-7}}. \quad (9)$$

The sign of $\sqrt{-7}$ depends on the choice of ε . If $\varepsilon = e^{2\pi i/7}$ is chosen, it is thereby fixed. The three numbers may also be written in the real form

$$\alpha = -\frac{2 \sin(8\pi/7)}{\sqrt{7}}, \quad \beta = -\frac{2 \sin(4\pi/7)}{\sqrt{7}}, \quad \gamma = -\frac{2 \sin(2\pi/7)}{\sqrt{7}}; \quad (10)$$

¹C. Jordan, Mémoire sur les équations différentielles linéaires à intégrale algébrique, Crelle's Journal für Mathematik, vol. 84 (1878); Sur la détermination des groupes d'ordre fini etc., Atti della R. Accademia di Napoli, vol. VIII (1879). Compare also Valentiner, Kjöbenhavn's Skrifter (6) V (1889).

they are the roots of

$$u^3 + u^2 - 1 = 0. \quad (11)$$

One also obtains the useful formulae

$$\alpha\varepsilon + \beta\varepsilon^{-2} + \gamma\varepsilon^2 = 1, \quad \alpha\varepsilon^{-1} + \beta\varepsilon^{-4} + \gamma\varepsilon^{-2} = 1, \quad (12)$$

together with the equations obtained from these by replacing ε by ε^{-1} .

With these determinations the substitutions τ, χ, ω satisfy the required relations. If, in addition,

$$\theta = \omega\tau\omega\tau^{-1}\omega, \quad (13)$$

then the group contains a substitution of seventh order which, together with the preceding elements, reproduces the relations of L_7 . Direct computation, using (6)–(12), gives in particular the comparison formulae

$$\begin{aligned} \alpha &= \beta^2\varepsilon^{-1} + \gamma^2\varepsilon^3 + \alpha^2\varepsilon^{-3} + \beta\gamma\varepsilon + \gamma\alpha + \alpha\beta\varepsilon, \\ \beta &= \gamma^2\varepsilon^{-2} + \alpha^2\varepsilon^3 + \beta^2\varepsilon^{-3} + \gamma\alpha\varepsilon^{-1} + \alpha\beta\varepsilon + \beta\gamma, \\ \gamma &= \alpha^2\varepsilon^{-4} + \beta^2\varepsilon^3 + \gamma^2\varepsilon^{-2} + \alpha\beta\varepsilon^{-2} + \beta\gamma\varepsilon^{-1} + \gamma\alpha, \end{aligned} \quad (14)$$

with the conjugate system obtained by $\varepsilon \mapsto \varepsilon^{-1}$. These formulae complete the construction of the ternary linear group of order 168.

§ 116. Poles and axes.

A ternary linear substitution

$$A = \begin{pmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{pmatrix} \quad (1)$$

associates with each point $(x_1 : x_2 : x_3)$ a point $(y_1 : y_2 : y_3)$. At the same time every straight line is carried into a straight line. The mapping of the plane into itself thus expressed is a collineation. If a second substitution S is used to change coordinates, then A goes over into the transformed substitution $S^{-1}AS$; transformation of the group is therefore equivalent to transformation of the coordinate system.

The points carried into themselves by A are determined from

$$\lambda x_1 = \alpha x_1 + \beta x_2 + \gamma x_3, \quad \lambda x_2 = \alpha' x_1 + \beta' x_2 + \gamma' x_3, \quad \lambda x_3 = \alpha'' x_1 + \beta'' x_2 + \gamma'' x_3. \quad (2)$$

Eliminating x_1, x_2, x_3 gives the cubic equation

$$\begin{vmatrix} \alpha - \lambda & \beta & \gamma \\ \alpha' & \beta' - \lambda & \gamma' \\ \alpha'' & \beta'' & \gamma'' - \lambda \end{vmatrix} = 0. \quad (3)$$

In general there are three points which are their own images. Weber calls them, as in the binary case, the poles of the substitution. The three joining lines of the poles are likewise their own images; they are the axes of the substitution.

Special cases may occur. Two or even three poles may coincide. More important is the case in which infinitely many points are their own images. Apart from multiplication, they then lie on one straight line. This line is called a principal axis of the substitution. A substitution with a principal axis has, besides it, one further pole, which in special cases may fall on the principal axis itself.

If we put the principal axis in the line $x_1 = 0$, then the condition that every point of this line remain fixed gives the normal form

$$y_1 = \alpha x_1, \quad y_2 = \beta x_2, \quad y_3 = \beta x_3 \quad (4)$$

for the essential part of the substitution. The lines through the pole not lying on the principal axis, and only those lines, remain fixed besides the principal axis. Thus the possible kinds of ternary linear substitutions are:

1. substitutions with three poles;
2. substitutions with two poles;
3. substitutions with one pole;
4. substitutions with a principal axis and a pole;
5. substitutions with a principal axis.

There is also, as a limiting case, multiplication, in which every point is its own image.

§ 117. Application to the group G_{168} . Sevenfold poles.

We now determine the poles and axes of the group G_{168} constructed in § 115. From the congruence group L_7 the orders and cycles of the group are already known. There are

48 elements of order seven, 56 elements of order three, 42 elements of order four, 21 elements of order two (1)

The elements of order seven, together with the identity, fall into eight cycles of seven terms; the elements of order three into 28 cycles of three terms; and the elements of order four and order two into 21 cycles of four terms.

If a point remains fixed under ν substitutions of the group, including the identity, we call it a ν -fold pole. For determining the poles it suffices to examine one representative of each cycle, since all the other cases are obtained by transformation.

For the substitution τ of order seven the vertices of the coordinate triangle are three distinct poles. All powers τ^k with exponent not divisible by 7 have the same three poles. The substitutions $\tau^r \chi^s$ carry these three poles cyclically into one another; the remaining substitutions carry the coordinate triangle into different coordinate triangles. Hence there are eight triples of such poles. Thus:

There are 24 sevenfold poles. They divide, corresponding to the eight seven-term cycles, into eight triples, and each one of the 24 points can be carried into any other by substitutions of the group.

§ 118. The principal axes.

We now consider the substitutions of order four and of order two. We choose ω and ω^2 as representatives. For the poles of ω one has the cubic equation

$$\begin{vmatrix} \gamma\varepsilon^2 - \lambda & \alpha\varepsilon^4 & \beta \\ \alpha & \beta\varepsilon^5 - \lambda & \gamma\varepsilon \\ \beta\varepsilon^{-2} & \gamma\varepsilon & \alpha - \lambda \end{vmatrix} = 0, \quad (1)$$

which, by the relations of § 115, reduces to

$$\lambda^3 - \lambda^2 + \lambda - 1 = 0. \quad (2)$$

The roots are $1, i, -i$. For ω^2 one obtains a double root $\lambda = -1$ and a simple root $\lambda = 1$. Substituting $\lambda = -1$, the three linear equations for the fixed point reduce to the single equation

$$\alpha\gamma x_1 + \beta\gamma\varepsilon^3 x_2 + \alpha\beta\varepsilon^5 x_3 = 0. \quad (3)$$

This equation represents a principal axis. By transformation it gives 21 different principal axes. If the corresponding linear forms are denoted by $A_{s,r}$, they may be written, up to the same notation as in the source, as

$$\begin{aligned} A_{1,r} &= \alpha\gamma\varepsilon^r x_1 + \beta\gamma\varepsilon^{3r+3} x_2 + \alpha\beta\varepsilon^{5r+5} x_3, \\ A_{2,r} &= \beta\gamma\varepsilon^{3r+1} x_1 + \alpha\beta\varepsilon^{5r+3} x_2 + \alpha\gamma\varepsilon^r x_3, \\ A_{3,r} &= \alpha\beta\varepsilon^{5r+2} x_1 + \alpha\gamma\varepsilon^r x_2 + \beta\gamma\varepsilon^{3r+4} x_3, \end{aligned} \quad (4)$$

where $r = 0, 1, \dots, 6$. It follows at once that these 21 axes are all distinct. Thus there are 21 principal axes, and each can be transformed into each of the others by the group.

Since there are only 21 principal axes, each remains fixed under eight substitutions. These eight substitutions form the group of the axis, denoted by G_A . Among them only the identity and one substitution of order two leave every point of the axis fixed; the others fix only two points on the axis. The isolated pole of ω^2 , for a suitable representative, has the coordinates

$$x_1 : x_2 : x_3 = \alpha\gamma\varepsilon^{-3} : \beta\gamma\varepsilon : \alpha\beta\varepsilon^5. \quad (5)$$

There are 21 such points. They are eightfold poles. Through each of these poles pass four principal axes.

The two other poles of the same substitution also lie on the principal axis, but are different from the eightfold poles just described. An arbitrary other point on the axis is carried into itself only by the identity and by one substitution of order two; it is therefore called a twofold pole. We obtain:

There are 21 eightfold poles and 42 fourfold poles. On each principal axis lie four eightfold and two fourfold poles. Through every eightfold pole pass four principal axes; every eightfold pole is transformable into every other eightfold pole, and every fourfold pole into every other fourfold pole.

§ 119. The threefold and sixfold poles.

We next consider the substitutions of third order, choosing as representative

$$Z = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}. \quad (1)$$

From $Z(x) = \lambda x$ it follows that $\lambda^3 = 1$. If ρ is an imaginary third root of unity, the three poles are

$$x_1 = x_2 = x_3, \quad x_1 = \rho x_2 = \rho^2 x_3, \quad x_1 = \rho^2 x_2 = \rho x_3. \quad (2)$$

There is, however, an essential difference among these three poles. The first remains fixed under ω , and is therefore at least sixfold; the other two are interchanged by ω .

Let these two latter points be denoted temporarily by π and π' . The substitutions fixing π form a subgroup that contains $1, Z, Z^2$. Since π is neither a sevenfold nor a fourfold pole and is not a pole of a substitution of order two, this stabilizer cannot have order greater than three. For the order 168 is not divisible by 9. Consequently:

There are 56 threefold poles. They belong in pairs to the same substitution and therefore form 28 pairs. Every such pole can be transformed into every other by substitutions of the group.

Put

$$T = x_1 + x_2 + x_3. \quad (3)$$

Then $T = 0$ is the joining line of the pair π, π' . This form is fixed by Z and changes sign under ω . Under the 28 substitutions $\omega^s \tau^r$ it gives 28 different linear forms, which may, up to a common factor, be written as

$$\begin{aligned} T_{2,r} &= \alpha^2 \varepsilon^{2+r} x_1 + \gamma^2 \varepsilon^{4+2r} x_2 + \beta^2 \varepsilon^{1+4r} x_3, \\ T_{3,r} &= \gamma^2 \varepsilon^{4+r} x_1 + \beta^2 \varepsilon^{1+2r} x_2 + \alpha^2 \varepsilon^{2+4r} x_3. \end{aligned} \quad (4)$$

On each of these lines lie three eightfold and two threefold poles.

The third pole of Z , the point $x_1 = x_2 = x_3$, remains fixed under the six substitutions

$$1, \quad Z, \quad Z^2, \quad \omega, \quad \omega Z, \quad \omega Z^2. \quad (5)$$

It cannot remain fixed under any further substitution of order two, for then it would have to lie on one further principal axis. But if one substitutes $x_1 = x_2 = x_3$ in the equations of the principal axes, each is satisfied for only one value of the parameter. Thus this pole is sixfold. There are 42 sixfold poles, all derived from one another by the group; through each pass three principal axes.

§ 120. The configuration of the group G_{168} .

The totality of straight lines and points considered in the preceding paragraphs is called the configuration of the group G_{168} . The whole system is carried into itself by the 168 substitutions of the group.

On every line A lie four points P_8 , and through every point P_8 pass four lines A . On every line T lie three points P_8 , and through every point P_6 pass three lines A . The points P_8 and P_6 together form the complete system of intersection points of the 21 lines A . Indeed, 21 lines meet pairwise in 210 points, and these points coalesce threefold or sixfold when three or four of the lines pass through one point; hence

$$210 = 21 \cdot 6 + 28 \cdot 3. \quad (1)$$

Since through every point P_6 pass three lines A , and since every line A must contain the same number of such points, there are four points P_6 on every line A . In the same way, through every point P_8 pass four lines T . None of the points P_6 lies on a line T .

Thus the systems of poles are

$$\begin{array}{ll} 21 & \text{eightfold poles } P_8, \\ 28 & \text{sixfold poles } P_6, \\ 42 & \text{fourfold poles } P_4, \\ 56 & \text{threefold poles } P_3, \\ 24 & \text{sevenfold poles } P_7, \\ & \text{infinitely many twofold poles } P_2. \end{array} \quad (2)$$

The twofold poles form a linear range on each principal axis; the other systems are finite orbits of the group. The configuration therefore contains not only the 21 principal axes and the 28 lines T , but also the joining lines of poles of the same type, in particular the lines through a P_6 and the two associated P_3 , and the 24 joining lines of corresponding P_7 , represented in the coordinate triangle by its sides.